GEK-83828A



**GE Drive Systems**

*Instructions*

# T-GSI
# PLANT COMPUTER DATALINK

# USER'S MANUAL

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# CHAPTER 1

## 1-1. INTRODUCTION

This publication describes the datalink control discipline for interfacing a customer's Plant Computer to the T-GSI system. The overall system comprises two functional parts--message level control discipline and the link level control discipline. The operation of each of these disciplines is transparent to the other. These components of the datalink design correspond to the higher level and lower level aspects of the plant datalink.

At the message level, the datalink works on a inquiry and response basis. The Plant Computer always initiates the inquiries and the T-GSI system generates the response.

Transparent to the message level control discipline, the link level control discipline defines the datalink traffic content. This lower level conforms to ANSI Standard X3.28 subcategory 3.1/F1.

The datalink supports full-duplex point-to-point communications. The datalink may be switched or unswitched.

The message level discipline is current of the publication cover date. However, GE reserves the right to implement modifications to the message level discipline at any time.

The hardware configuration is illustrated in Figure 1.

**** NOTE ****

Numbers throughout this publication are expressed in either decimal or hexadecimal notation. The letter "D" following a number denotes decimal notation; the letter "H" denotes hexadecimal notation.

## 1-2. APPLICABLE DOCUMENTS

A detailed description of the electrical interface and the link level line control discipline are provided by the following documents:

1. ANSI Standard X3.28-1976
2. EIA Standard RS-232C

## 1-3. MESSAGE LEVEL CONTROL DISCIPLINE

The message control level discipline describes the procedures for transferring process related information between the Plant Computer and the T-GSI system.

The message level control discipline allows the Plant Computer to:

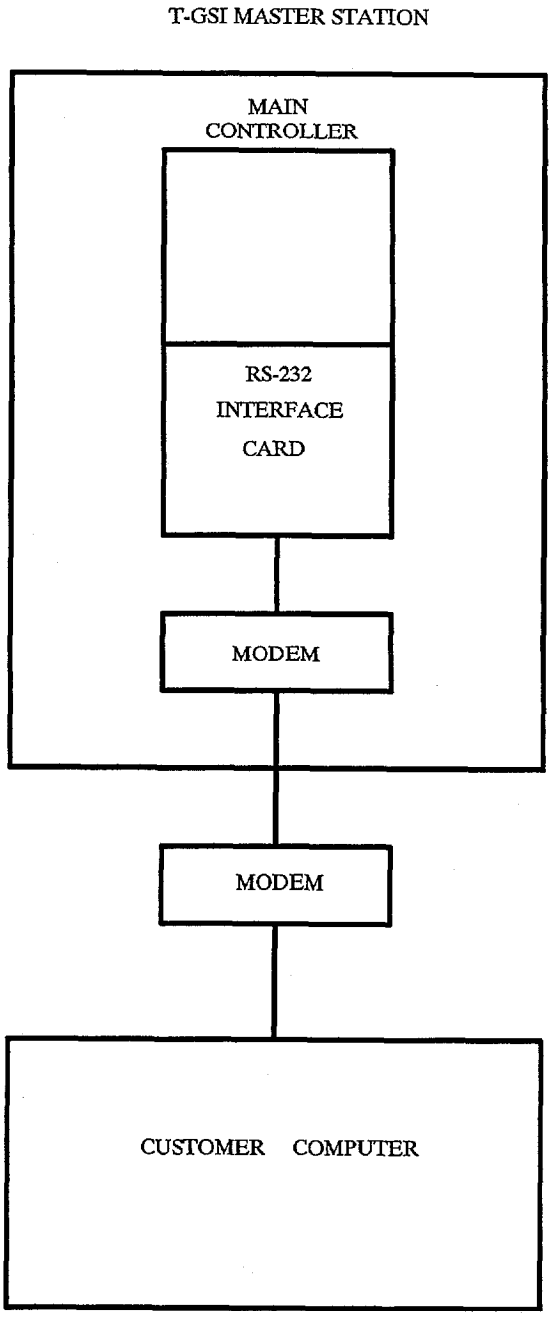



**Figure 1. System Configuration Diagram**

* Obtain the current status of all T-GSI inputs.
* Obtain T-GSI system status.
* Obtain T-GSI display parameters.

To perform any of the above functions, the Plant Computer issues a two-byte command message to the T-GSI System. The T-GSI System decodes the request, builds the response message per the command message, and transmits the response message. If an invalid command message is received, the first byte received, followed by a ASCII "?", is transmitted back to the Plant Computer.

Because the Plant Computer can issue commands at a faster rate than the T-GSI System can respond, the Plant Computer inquiries are queued by the T-GSI system. A maximum of one outstanding inquiry can be handled by the T-GSI system. If the limit is exceeded, the succeeding inquiries are ignored.

The messages passed between the Plant Computer and the T-GSI system are contained within the data field area of the link level control discipline, which is transparent to the message level control discipline.

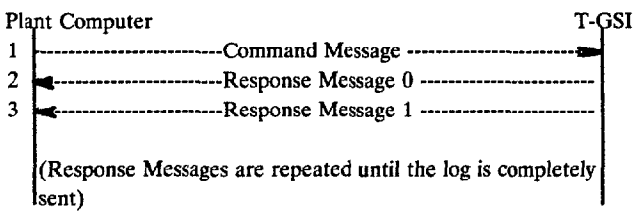
The following descriptions detail the inquiry-response sequence that occurs for a given Plant Computer to T-GSI system operation.

## 1-3.1. TO OBTAIN A FULL T-GSI SYSTEM LOG

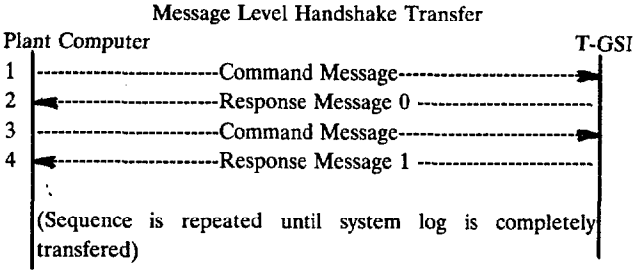
The Plant Computer can obtain a system log by transmitting the system log command message to the T-GSI system. The T-GSI system responds, on a time available basis, with the status of all active T-GSI channels. This response message is called a complete system log. Unused channel data is not sent. The representation of the channel values (as either hexadecimal or decimal ASCII codes) and the format of the system log is user-definable via the command option field, described below.

The Plant Computer specifies the content and format of a system log through the use of the Option field of the Command Message. The data can be transmitted in a decimal ASCII mode (directly readable) or in a Hex-ASCII mode (corresponding to the internal T-GSI representation).

It may be necessary for the T-GSI system to send a system log in several Response Messages. Such a situation is illustrated below.




When multiple transmissions are required to fetch a system log, the Plant Computer may wish to obtain a system log in a "stop-and-go" mode. In this mode of operation, called Message Level Handshake, the T-GSI system will only send a single Response Message for each Plant Computer Command Message. This situation is illustrated below.




## 1-3.2. Command Message Format

The command message format is shown below:

| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | Command Op Code |
| 0 | 1 | L | M | 0 | C | H | A | Command Options |

**1-3.2.1. Command Op Code.** The Command Op Code is the value sent by the Plant Computer to request a new system log from the T-GSI system. This will always be 40 hexidecimal. If the Response Message is greater than 128 bytes, the Plant Computer may request that the Response message be transmitted in blocks and at a rate determined by the Plant Computer. This mode called Message Level Handshaking permits the Plant Computer to limit the number of bytes of data that can be sent by the T-GSI system in one transmission.

To fetch a complete response, the Plant Computer must issue a command message for each block of data. The number of responses required to fetch a complete system is determined by the command options enabled and the number of channels in the system.

When the Message Level Handshake option is selected, the Plant Computer must successively send command messages (with the M field of the Command Option set) until the last response message is sent. The Inquiry Status field (See R field) of the response message will flag the last block of the system log.

**1-3.2.2. Command Option.** The Command Option field permits the Plant Computer to define the contents of a T-GSI system log. The options are selected by setting the respective option bit to a "1". Each option is independent of the other options. If no option is

selected, the T-GSI system will respond with the time/date and turbine number.

Field Description

A When set, causes channel values to be logged in a decimal ASCII format using eight ASCII characters. Channels are logged in ascending channel number order.

If the channel has "failed", denoted by an "F" in the alarm status field, no channel value is returned.

The layout of the values is as follows:

ASCII Format (analog value):

```
SXXXX. C
||__||||
| |  ||+--- Channel alarm status
| |  ||       "N" = Normal condition
| |  ||       "F" = Failed channel
| |  ||       "A" = Red alarm condition
| |  ||       "P" = Yellow alarm condition
| |  ||             (if applicable)
| |  ||
| |  ||
| |  |+------ Field delimiter (ASCII " ")
| |  |
| |  +------- Decimal point (ASCII ".")
| |           appropriately adjusted
| +---------- Channel value (0 to 9999)
|             right justified and
|             blank filled
+------------ Value polarity
               "-" = negative value
               " " = positive value
```

ASCII Format (digital value):

```
YYYYYY C
|____| |
  |    +----  Channel alarm status
  |            "N" = Normal condition
  |            "F" = Failed channel
  |            "A" = Alarm condition
  |            "P" = Pre-alarm condition
  |
  +------  Channel value
           "TTTTTT" = True/Logic "1"
           "FFFFFF" = False/Logic "0"
```

Field Description

H This field, when set, results in a Hex-ASCII dump of the internal T-GSI representation of a system log. Four ASCII characters, corresponding to the 16-bit T-GSI representation, are returned to the Plant Computer. NOTE: Only channels present in the system log are sent. To interpret a hex-ASCII dump, the user must use the Raw Data Descriptor database (See "Obtain T-GSI Raw Data Description").

Hex-ASCII Format (analog or digital value):

```
ZZZZ
 +---- Channel value (Hex-ASCII representation)
```

NOTE: Hex-ASCII representation will encode the 16-bit channel value into a four-character ASCII string. For example, the hexadecimal channel value "1 2 3 B" is encoded as the hex-ASCII string "31 32 33 42".

Channel information is stored internally in one of two formats, depending on whether it represents analog or digital data:

Analog Input

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| signed two's complement representation | | | | | | | | | | | | | | S | S |

Digital Input

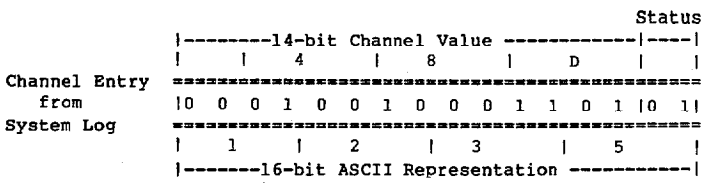
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| all 1's=true or all O's=false | | | | | | | | | | | | | | S | S |

| S | S | Description |
|---|---|---|
| 0 | 0 | Normal |
| 0 | 1 | Pre-alarm condition |
| 1 | 0 | Alarm condition |
| 1 | 1 | Failed channel |

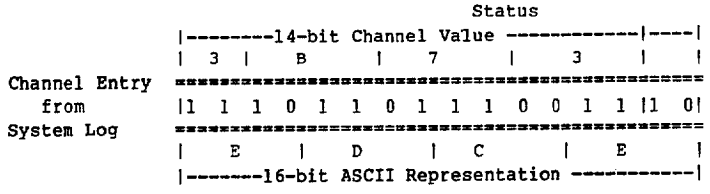
As shown above, the channel value comprises two fields a value field (bits 2 - 15) and a status field (bits 0 and 1). The channel value is a 14-bit, two's complement (signed) binary number. The two-bit status field defines the operational status of the channel.

The following two examples illustrate how the T-GSI system stores the channel value and status. Note that the position of the decimal point is not given in the representation. To determine the decimal point position see "Obtain T-GSI Raw Data Descriptors".

The first example depicts a channel value of 1165D (48DH) and a pre-alarm status:




The second example depicts a channel value of -1165D (3B73H) and an alarm condition:




Field Description

C This field will cause the channel number to be returned along with the corresponding value. The channel number will be represented by four ASCII characters consisting of three numeric characters and followed by a blank character. This field will precede the channel value.

The format is illustrated below.

```
CCCb
|_||
 | |
 | +----- Field delimiter (blank char.)
 |
 +------- Channel number (blank filled and
          right justified)
```

Field Description

M The Message Level Handshake option inhibits the T-GSI system from sending a response message until a command message is received. Since a system log requires several response messages to transfer the complete log, the M option allows the Plant Computer to control the rate at which the T-GSI transfers the system log. If this option is not selected, the T-GSI system will transmit all response messages required to send the complete log upon receiving the first command message.

Field Description

L The Auto Carriage Return/Linefeed option will cause the information associated with a channel to be terminated with ASCII carriage return and linefeed characters.

### 1-3.3. Response Message Format

The layout of the maximum-case system log request (i.e. all Command options selected), as sent by the T-GSI system, is illustrated in Figure 2. The message is depicted as it would be seen in the Plant Computer's input buffer.

**1-3.3.1. Response Header.** The Response Header returned to the Plant Computer from the T-GSI system is identical with the Command Message Op Code.

**1-3.3.2. Inquiry Status.** When set, the one-bit R field indicates the following message completes the log request. This field is associated with the Message Level Handshake option.

R Description

1 Last response message

0 Intermediate response message (Succeeding response message(s) required to complete system log)

The A field is a four-bit field indicating if the request was recognized. If an error is detected by the T-GSI system, resulting in a 3FH (ASCII "?") inquiry status and a carriage return-line feed delimiter, then log information will not be returned in the Response Message.

| A A A A | Description |
|---|---|
| 0 0 0 0 | No errors |
| 0 0 0 1 to 1 1 1 0 | (Not used) |
| 1 1 1 1 | Inquiry message error |

**1-3.3.3. Log Date.** The Log Date is an eight-character numeric ASCII string (including "/" separators) defining the date when the returned log was taken. The format of the date is "Y2Yl/M2Ml/D2Dl".

**1-3.3.4. Log Time.** The Log Time is an eight-character numeric ASCII string (including ":" separators) defining the time when the returned log was taken. The format of the time is "h2hl:m2ml:s2sl". Time is given in terms of a 24-hour clock.

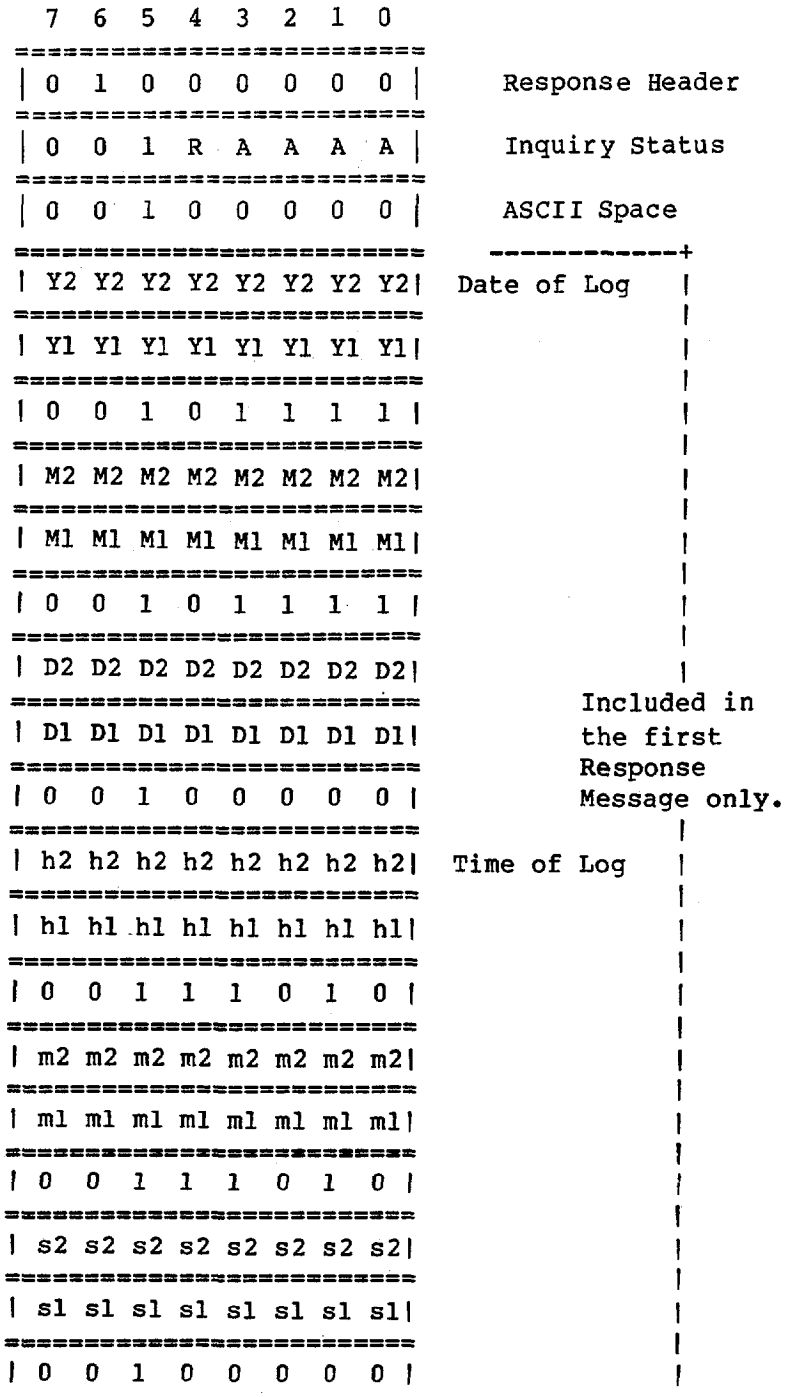



**Figure 2. Response Message Format (System Log Request)**
**(Sheet 1 of 3)**

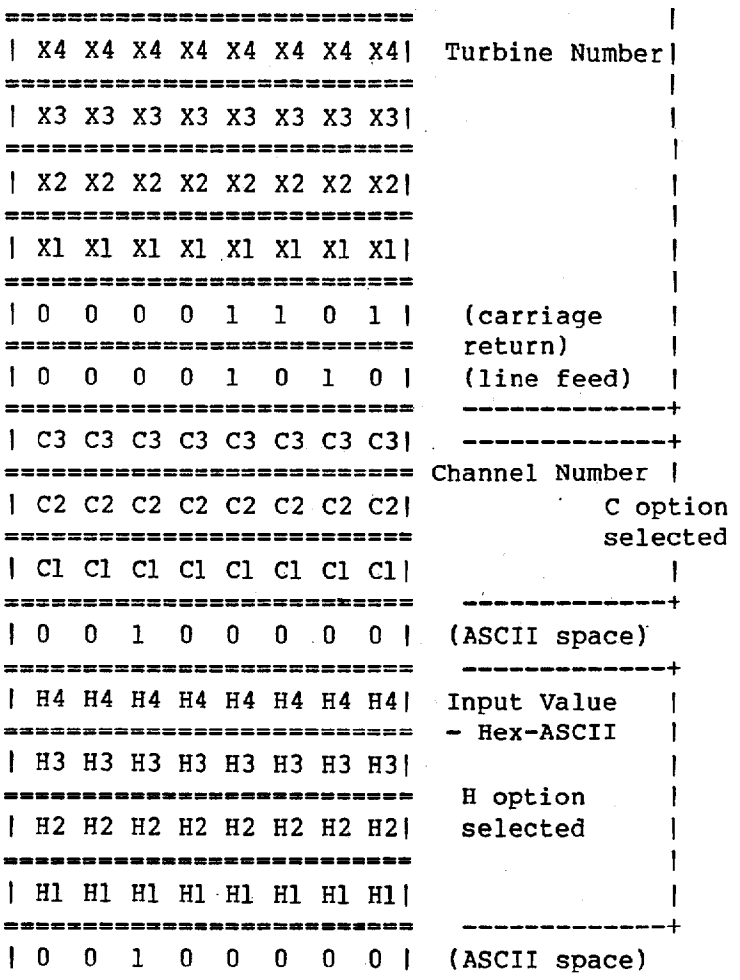



**Figure 2A. Response Message Format (System Log Request)**
**(Sheet 2 of 3)**

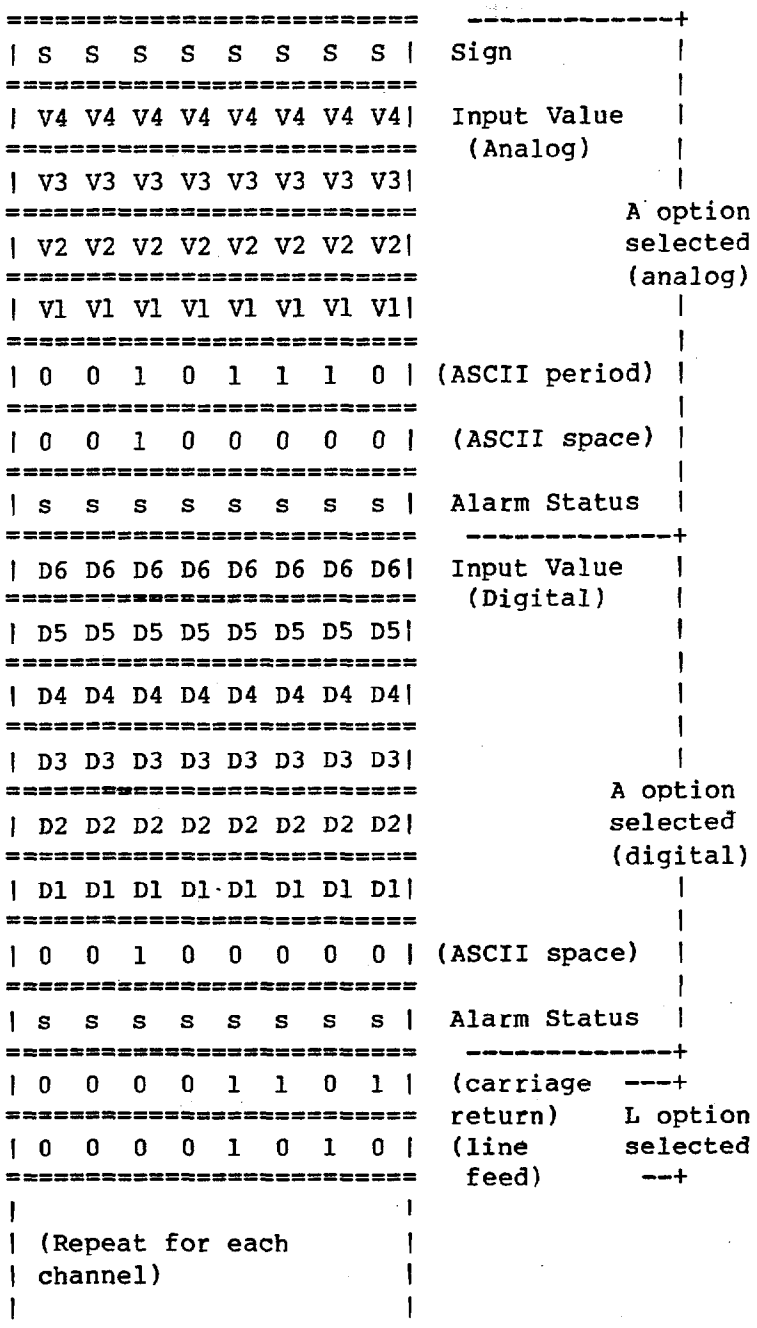



**Figure 2B. Response Message Format (System Log Request)**
**(Sheet 3 of 3)**

**1-3.3.5. Turbine Number.** The Turbine Number, fields XI to X4, is a four-character alphanumeric ASCII string corresponding to the system turbine number.

**1-3.3.6. Channel Number.** The Channel Number, fields C3 to Cl, is a three-character numeric ASCII string specifying the channel number associated with the channel input value field. An ASCII space character is used to delimit the Channel Number from the Input Value.

**1-3.3.7. Input Value - H option.** The Input Value (H option selected), fields H4 to Hl, is the Hex-ASCII representation corresponding to the internal T-GSI 16-bit hexadecimal representation of a channel value.

It should be noted that channel numbers, signal polarity, and decimal point position are not included with the value.

The Plant Computer must format the channel values. All information needed to format the internal T-GSI representation can be obtained from the Raw Data Descriptor Database.

**1-3.3.8. Input Value - A Option.** The Input Value (A Option selected), fields S to V4 to Vl to s, is an eight-character alpha-numeric ASCII string, in base 10 with sign and decimal point, of the channel value.

The Sign is a single ASCII character corresponding to the sign of the Input Value. If the Input Value is positive, the Sign will be an ASCII space. If the Input Value is negative, the Sign will be an ASCII dash character (2D hexadecimal).

The Input Value is a string of five ASCII characters corresponding to the value of the input. If the Input Value represents analog data, the string will be four ASCII numeric characters and an ASCII period character (appropriately positioned). If the Input Value represents digital data, the string will be six ASCII "T"'s or "F"'s corresponding to true and false states respectively.

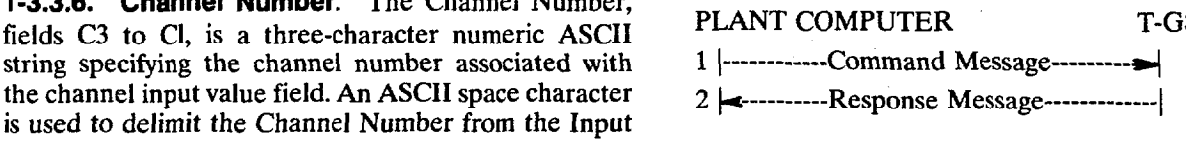
The Input value is delimited from the Alarm Status by an ASCII space character.

The Alarm Input Status is a single ASCII character corresponding to the operational state of the input. The ASCII letter will be interpreted as follows:

| Input Status | Description |
|---|---|
| N | No error |
| P | Pre-alarm condition |
| A | Alarm condition |
| F | Failed input |

## 1-4. TO OBTAIN T-GSI SYSTEM STATUS

The system status command permits the Plant Computer to read the current operational state of the T-GSI system.




### 1-4.1. Command Message Format

The command message format is shown below:

| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | Command Op Code |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | Command Option |

**1-4.1.1. Command Op Code.** The Command Op Code is the value sent by the Plant Computer to request system status from the T-GSI system. This will always be 42 hexadecimal.

**1-4.1.2. Command Option.** The Command Option will always be 30 hexadecimal.

### 1-4.2. Response Message Format

The response message format is illustrated in Figure 3.

**1-4.2.1. Response Header.** The Response Header returned to the Plant Computer from the T-GSI system for the system status command will be 42 hexadecimal.

**1-4.2.2. Inquiry Status.** The four-bit A field indicates if the request was recognized.

| A A A A | Description |
|---|---|
| 0 0 0 0 | No errors. |
| 0 0 0 1 to 1 1 1 0 | (not used) |
| 1 1 1 1 | Inquiry message error. |

**1-4.2.3. Status Date.** The Status Response Date is an eight-character numeric ASCII string (including "/" separators) defining the date when the returned Status Response was taken. The format of the date is "Y2Yl/M2Ml/D2Dl".

**1-4.2.4. Status Response Time.** The Status Response Time is an eight-character numeric ASCII string (including ":" separators) defining the time when the

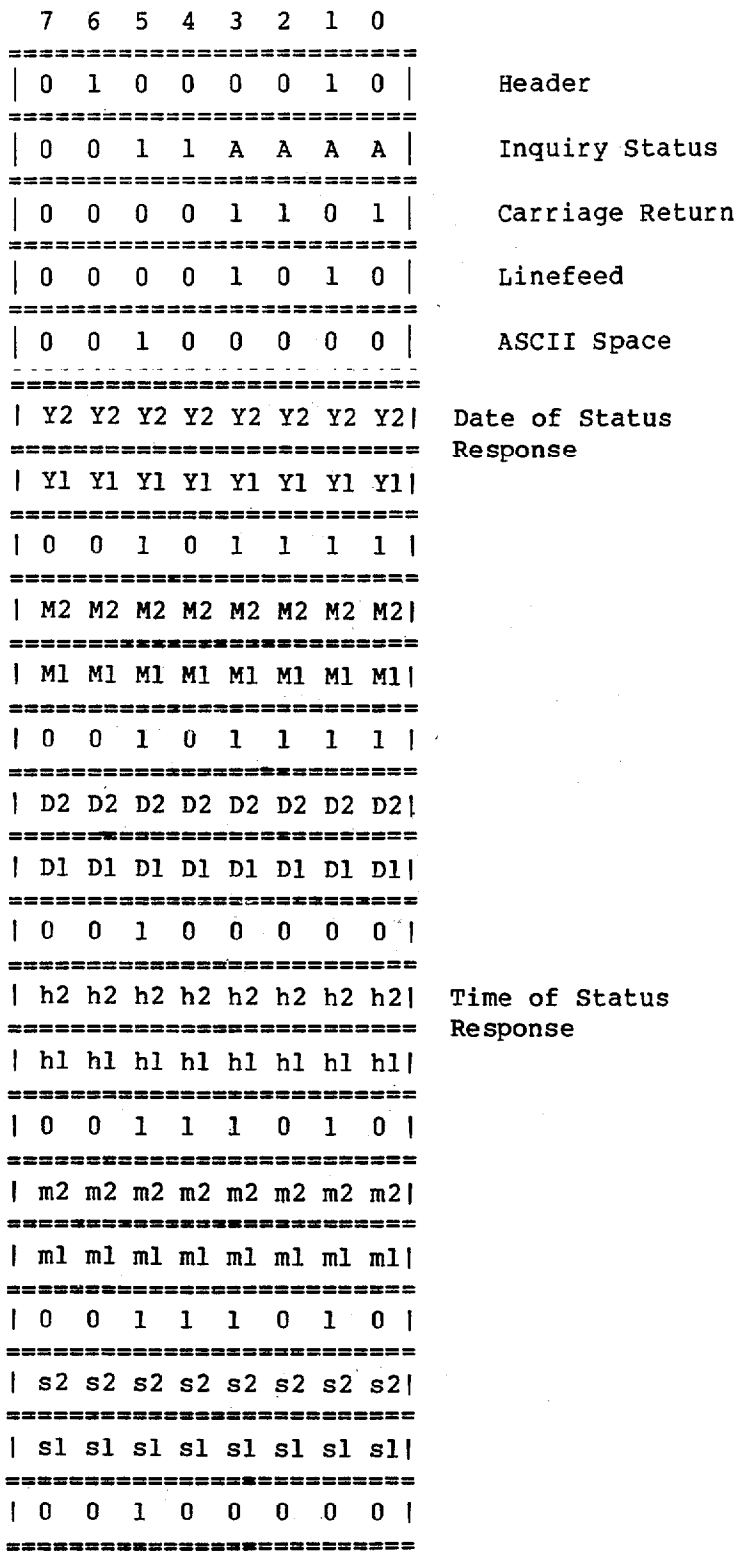



**Figure 3. Response Message Format (System status)**
**(Sheet 1 of 3)**

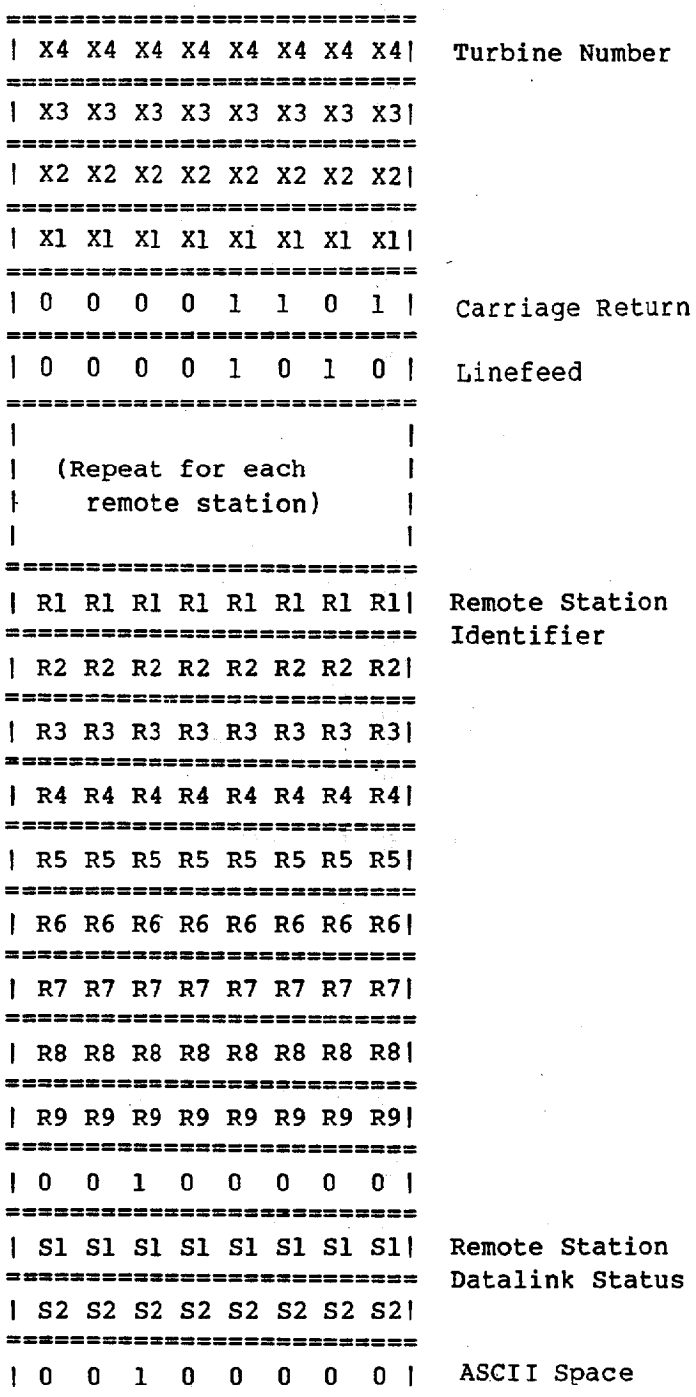



**Figure 3A. Response Message Format (System Status)**
**(Sheet 2 of 3)**

| | |
|---|---|
| \| E3 E3 E3 E3 E3 E3 E3 E3\| | Remote Station Datalink Error Count |
| \| E2 E2 E2 E2 E2 E2 E2 E2\| | |
| \| E1 E1 E1 E1 E1 E1 E1 E1\| | |
| \| 0 0 0 0 1 1 0 1 \| | Carriage Return |
| \| 0 0 0 0 1 0 1 0 \| | Linefeed |
| \| P1 P1 P1 P1 P1 P1 P1 P1\| | Plant Computer Identifier |
| \| P2 P2 P2 P2 P2 P2 P2 P2\| | |
| \| P3 P3 P3 P3 P3 P3 P3 P3\| | |
| \| P4 P4 P4 P4 P4 P4 P4 P4\| | |
| \| P5 P5 P5 P5 P5 P5 P5 P5\| | |
| \| P6 P6 P6 P6 P6 P6 P6 P6\| | |
| \| P7 P7 P7 P7 P7 P7 P7 P7\| | |
| \| P8 P8 P8 P8 P8 P8 P8 P8\| | |
| \| P9 P9 P9 P9 P9 P9 P9 P9\| | |
| \| 0 0 1 0 0 0 0 0 \| | ASCII Space |
| \| 0 1 0 1 0 1 0 1 \| | |
| \| 0 1 0 1 0 0 0 0 \| | |
| \| 0 0 1 0 0 0 0 0 \| | |
| \| E3 E3 E3 E3 E3 E3 E3 E3\| | Plant Computer Datalink error count |
| \| E2 E2 E2 E2 E2 E2 E2 E2\| | |
| \| E1 E1 E1 E1 E1 E1 E1 E1\| | |
| \| 0 0 0 0 1 0 1 0 \| | |
| \| 0 0 0 0 1 1 0 1 \| | |

**Figure 3B. Ressponse Message Format (System Status)**
**(Sheet 3 of 3)**

returned Status Response was taken. The format of the time is "h2hl:m2ml:s2sl". Time is given in terms of a 24 hour clock.

**1-4.2.5. Turbine Number.** The Turbine Number, fields Xl to X4, is a four-character alphanumeric ASCII string corresponding to the system turbine number.

**1-4.2.6. Remote Station Identifier.** The Remote Station Identifier fields Rl to R9 is a nine-character alphabetic ASCII string specifying a remote station. The string preceeds the datalink status and datalink error count associated with the remote station.

**1-4.2.7. Remote Station Datalink Status.** The Remote Station Datalink Status (fields Sl to S2) is a two-character alphabetic ASCII string defining the operational status of the remote station specified in fields Rl to R9. The status will either be "UP" datalink operational or "DN" datalink is non-operational.

**1-4.2.8. Remote Station Datalink Error Count.** The Remote Station Datalink Error Count (fields El to E3) is a three character numeric ASCII string specifying the number of detected remote station datalink errors. The value will range between "0" and "255". The error count will saturate (stop) at "255" until manually reset.

**1-4.2.9. Plant Computer Datalink Identifier.** The Plant Computer Datalink Identifier (fields Pl to P9) is a nine-character alphabetic ASCII string identifying the Plant Computer. The character string preceeds the datalink error count. The Status field will always be "UP" since it is implicit that datalink status is operational.

**1-4.2.10. Plant Computer Datalink Error Count.** The Plant Computer Datalink Error (fields El to E3) is a three character numeric ASCII string defining the number of detected Plant Computer datalink error counts. The count is the sum of transmit and receive errors. The error count will range between "0" and "511". The count will saturate at "511" until manually reset.

## 1-5. TO OBTAIN T-GSI RAW DATA DESCRIPTORS

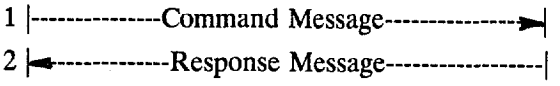
The command described below permits the Plant Computer to obtain the T-GSI system's Raw Data Descriptor records. This database is necessary to interpret system logs obtained using the Hex-ASCII option (See Command Option - H field). Specifically, the database defines:

- The starting channel number and the number of channels defined by a Raw Data Descriptor Record; and
- The implied decimal point position for a channel obtained with the Hex-ASCII option and the signal class -- analog or digital.

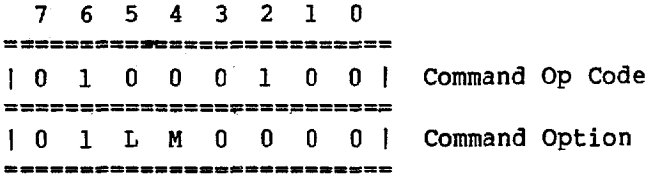
Using this information the user can associate channel numbers, signal class, and decimal point position to a Hex-ASCII dump of the system log.

The Raw Data Descriptor database is fixed and describes only the channels in the system. Therefore, the Plant Computer needs to execute this Command Message only once.

The protocol to obtain the Raw Data Descriptor database, illustrated below, is identical to previous Plant Computer - T-GSI interchanges.




### 1-5.1. Command Message Format




**1-5.1.1. Command Op Code.** The Command Op Code is the value sent by the Plant Computer to request the T-GSI system for T-GSI raw data descriptors. This will always be 44 hexadecimal.

**1-5.1.2. Command Option.** The Command Option has two fields. When invoked, the "L"option will cause ASCII carriage return/linefeed terminators to be used as record separators.

| Field | Description |
|---|---|
| L | This field permits the Plant Computer to require ASCII carriage return/line feeds between Response Message records. |

| Field | Description |
|---|---|
| M | The Message Level Handshake option causes T-GSI system to transmit data, which requires multiple Response Message(s) to be sent, to inhibit transmitting suceeding Response Message(s) until another Command Message is is- |

sued. If the option is not selected the T-GSI system will transmit all Response Messages required to send the complete data -- as initiated by one Command Message.

### 1-5.2. Response Message Format

The response message format is shown in Figure 4.

**1-5.2.1. Response Header.** The Response Header returned to the Plant Computer from the T-GSI system for this command will be 44 hexadecimal.

**1-5.2.2. Inquiry Status.** When set, the one-bit R field indicates the following message completes the log request. This field is associated with the Message Level Handshake option.

| R | Description |
|---|---|
| 1 | Last response message |
| 0 | Intermediate response message (Succeeding response message(s) required to complete system log) |

The four-bit A field indicates if the request was recognized.

| A A A A | Description |
|---|---|
| 0 0 0 0 | No errors. |
| 0 0 0 1 to 1 1 1 0 | (not used) |
| 1 1 1 1 | Inquiry message error. |

**1-5.2.3. Raw Data Descriptor Date.** The Raw Data Descriptor Date is an eight-character numeric ASCII string (including "/" separators) defining the date when the returned Raw Data Descriptor was taken. The format of the date is "Y2Yl/M2Ml/D2Dl".

**1-5.2.4. Raw Data Descriptor Time.** The Raw Data Descriptor Time is an eight-character numeric ASCII string (including ":" separators) defining the time when the returned Raw Data Descriptor was taken. The format of the time is "h2hl:m2ml:s2sl". Time is given in terms of a 24-hour clock.

**1-5.2.5. Turbine Number.** The Turbine Number, fields Xl to X4, is a four-character alphanumeric ASCII string corresponding to the system turbine number.

**1-5.2.6. Raw Data Descriptor Record.** The Raw Data Descriptor Record is a fixed length record which describes the common display attributes of a group of channels. The order of the records corresponds to the order of the channels in the system log.

There are four fields of interest. They are:

| | |
|---|---|
| SCN | Starting Channel Number |
| NOCH | Number of Channels |
| LOG | Signal Class |
| LDP | Number of Digits to the Left of the Decimal Point. |

A Hex-ASCII dump of the system log sends only the channels present in the system and does not send channel numbers. As a result, to determine the channel number of an entry in the system log, the user must compute the channel based on the SCN and NOCH fields of the Raw Data Descriptor database. Futhermore, determination of the channel number for a given system log entry requires the user to begin with the first Raw Data Descriptor and serially index through the records using the SCN and NOCH information until the record covering the system log entry is found.

The identical process must be performed to obtain the signal class and decimal point positioning information.

The layout of one Raw Data Descriptor Record is illustrated in Figure 5.

As an example, the following RDDB defines channels 11 to 19 as analog channels where there are three decimal digits to the left of the decimal point:

| Byte | Value | Comments |
|---|---|---|
| 0 | 0B | Starting channel is 11. |
| 1 | 00 | |
| 2 | 09 | There are 9 channels. |
| 3 | XX | Note: XX = Don't Care |
| 4 | XX | |
| 5 | XX | |
| 6 | XX | |
| 7 | XX | |
| 8 | XX | |
| 9 | XX | |
| 10 | XX | |
| 11 | XX | |
| 12 | XX | |
| 13 | 10 | NOTE: Bits 0:2 and Bit 6 are Don't Care. |
| 14 | XX | |
| 15 | XX | |

If the Hex-ASCII value of channel number 13 were 1236H, then the display value of the channel is +116.5 in the abnormal condition.

This example indicates that the first nine channel values (the number of channels is indicated by byte 2) represent data from channels 11 thru 19 (Bytes 1 and 0 indicate the channel number to be assigned to the first channel record).

The tenth and subsequent channel values must be interpreted using the next successive Raw Data Descriptor Field.

## 1-6. LINK LEVEL CONTROL DISCIPLINE

The Link Level Control Discipline defines the procedures for the Plant Computer and the T-GSI system to

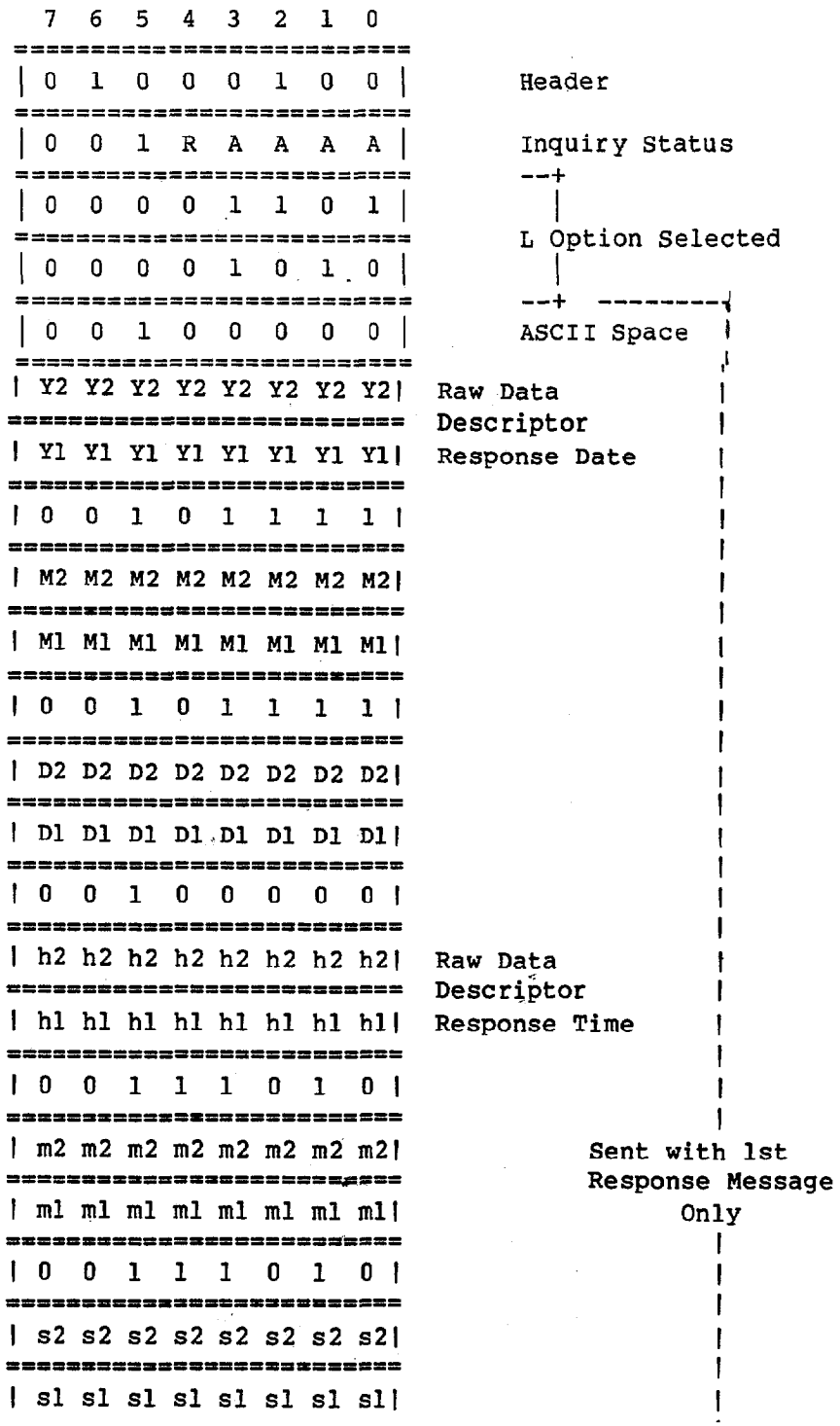



**Figure 4. Response Message Format (Raw Data Descriptors) (Sheet 1 of 2)**

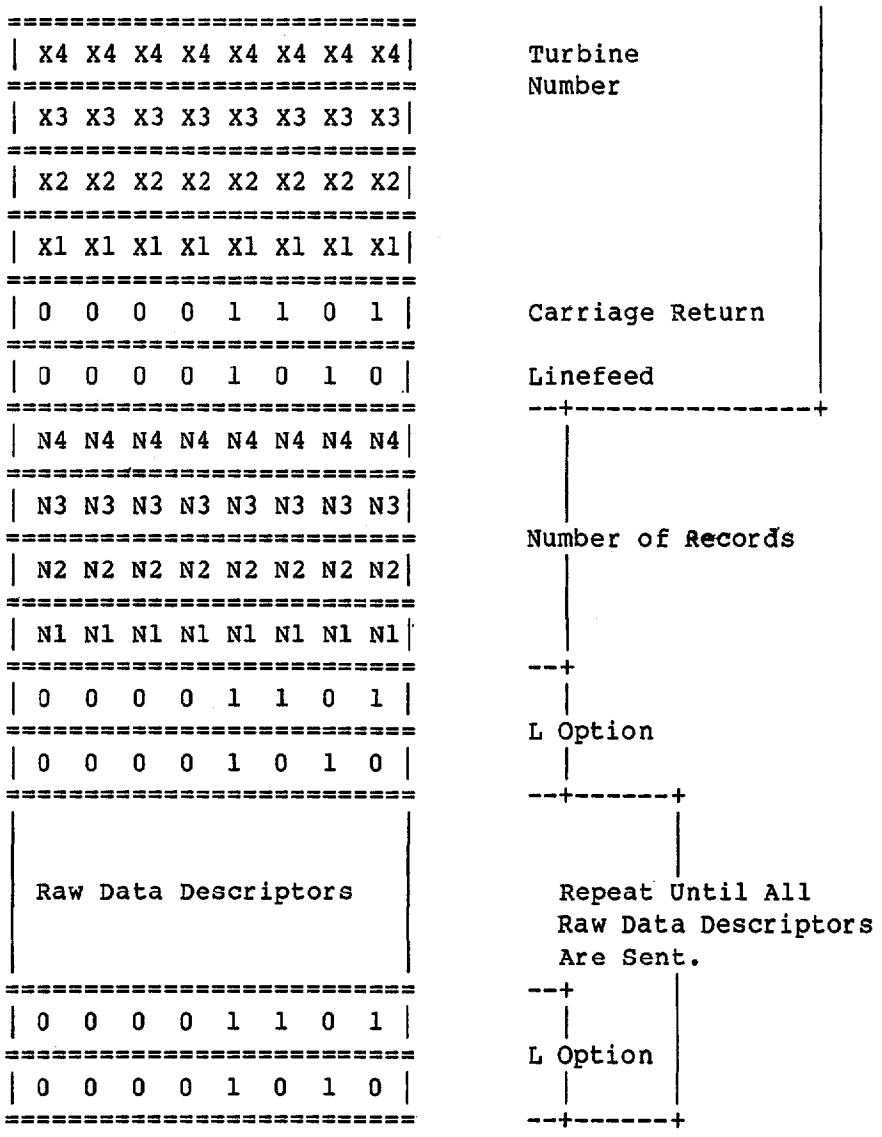
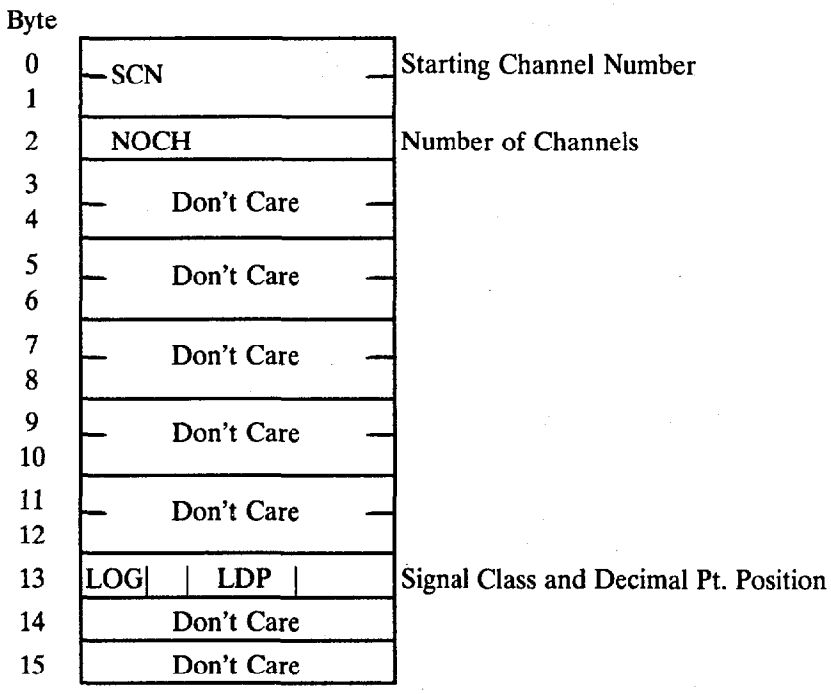



**Figure 4A. Response Message Format (Raw Data Descriptors)**
**(Sheet 2 of 2)**




| Field | Byte | Bits | Description |
|---|---|---|---|
| SCN | 0 | 7:0 | Starting Channel Number - Low Half. |
| | 1 | 7:0 | Starting Channel Number - High Half. |
| NOCH | 2 | 7:0 | Number of Channels - Starting with SCN |
| LOG | 13 | 7 | Signal Class (O = Analog\|l = Digital) |
| LDP | 13 | 5:3 | Number Digits to Left of Decimal Point |

Figure 5. Raw Descriptor Record

communicate on a byte by byte basis. The protocol for interchanging data and control is defined by ANSI Standard X3.28, subcategory 3.1/Fl.

The salient features of this protocol are:

- Two way simultaneous communications.
- Point-to-point full duplex communications.
- Switched or unswitched communications.
- Embedded message responses.
- Block orientated messages.

Message headers must be sent to the T-GSI system, where the header prefix will be a three character ASCII sequence: "TGI". The T-GSI will respond to the Plant Computer with a header and the same header prefix.

The T-GSI system will execute a link terminate sequence if a T-GSI to Plant Computer transmission fails. Reasons for a link failure would be either an invalid reply or no reply.

The T-GSI system has fully implemented the protocol with the exception of the following points.

1. The Block Abort procedure is not supported (See Section 3.4.1 of ANSI Standard).
2. The Sending Station Abort procedure is not supported (See Section 3.4.2 of ANSI Standard).
3. The Termination Interrupt procedure is not supported (See Section 3.4.3 of ANSI Standard).
4. The Reverse Interrupt procedure is not supported (See Section 3.4.4 of ANSI Standard).
5. The WACK sequence is not supported (See Section 3.4.5 of ANSI Standard ).
6. The Timer C function is not implemented (See Section 3.5.2.3 of ANSI Standard).
7. The Timer D function is not implemented (See Section 3.5.2.4 of ANSI Standard).
8. The Reply-Request Supervisory sequence is not supported (See Section 3.14.3 of ANSI Standard).
9. Data transparency is not implemented (See Section 3.4.1.4 of the ANSI Standard).

The T-GSI system has implemented the protocol with the following parameters:

1. Data block size = 128 bytes
2. Message prefix = "TGI"
3. Message header = "TGI"
4. Timer A (Response Timer) =

$$\left[\frac{\text{(No. Bytes/Data Block) * (No. bits/Character)}}{\text{Datalink Baud rate}} + 1\right] \text{Seconds}$$

5. Timer B (Receive Timer) = 8 minutes
6. Number of message retries = 2

## 1-7. ELECTRICAL INTERFACE

A RS-232C hardware interface is used by the T-GSI system. A modem is supplied with the system. The following parameters apply to the interface.

1. Full duplex transmissions
2. Asynchronous communication
3. Transmission rate = 9600 bits per second
4. Parity = even
5. Number of bits per character = 7
6. Number of stop bits = 1

## 1-8. PERFORMANCE SPECIFICATIONS

The following timing estimates are based on a 9600 BAUD datalink. It is further assumed that the Plant Computer command requests are made with the datalink already established and that there are no link level retries.

### 1-8.1. OBTAIN T-GSI SYSTEM LOG TIMING

The data throughput of the datalink for a T-GSI System Log Request is determined by two factors:

1. Number of channels passed by the T-GSI System and the requested format (ASCII or Hex ASCII); and
2. The T-GSI System response time to service a Plant Computer's request.

The number of channels contained with a log is fixed. Only those channels implemented by system are transmitted. Unused channels are not sent.

To determine the total time to obtain a T-GSI Log, use the table below.

| | T-GSI RESPONSE REQUEST | TIME TO SERVICE RESPONSE MESSAGE TIME |
|---|---|---|
| ASCII FORMAT TIME | 3 sec max | 52ms + ((4.7ms) * (No. Channels - 7)) + (4.7ms * (Integer ((No. Channels-7)/8) + 1)) |
| HEX ASCII FORMAT TIME | 3 sec max | 52ms + ((1.3ms) * (No. Channels - 25)) + 1.3ms * (Integer ((No. Channels - 25)/31) + 1) |

ms = .001 seconds

The times include the time from the first byte of the Plant Computer request until the last byte of the T-GSI Response message.

## 1-8.2. OBTAIN T-GSI SYSTEM STATUS TIMING

The time delay for a T-GSI System Status request is defined as follows:

| | T-GSI RESPONSE TIME TO SERVICE REQUEST | RESPONSE MESSAGE TIME |
|---|---|---|
| T-GSI SYSTEM STATUS TIME | 3 sec max | 39 ms |

## 1-8.3. OBTAIN T-GSI RAW DATA DESCRIPTOR TIMING

The time required to send the T-GSI Raw Data Descriptors will vary according to the system and directly corresponds to the number of descriptors in the system. The time delay to service a T-GSI Raw Data Descriptors is as follows:

| | T-GSI RESPONSE TIME TO SERVICE REQUEST | RESPONSE MESSAGE TIME |
|---|---|---|
| T-GSI RAW DATA DESCRIPTOR TIME | 3 sec max | 24ms + (13ms * number of Raw Data Descriptors) |

A typical T-GSI system will have 75 Raw Data Descriptors. Thus, the time required to send the response message would be 999ms.

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